

Roll No.

TMA-316

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
DISCRETE STRUCTURES AND COMBINATORICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Prove that :

$$A - (B \cap C) = (A - B) \cup (A - C).$$

OR

(b) $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = ax + b$, where $a, b, x \in \mathbb{R}$ and $a \neq 0$.

Show that f is invertible and find the inverse of f . (CO1)

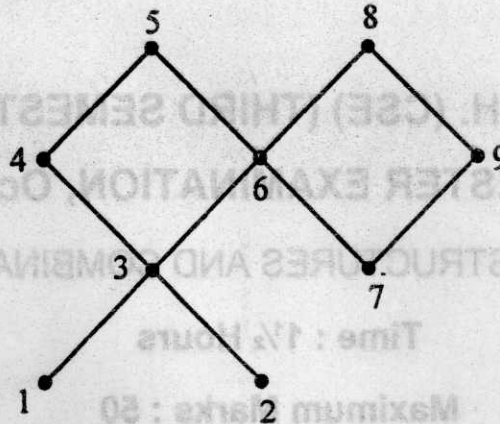
2. (a) The relation on a set $A = \{1, 2, 3, 4\}$ is defined by $R = \{(1, 1), (1, 2), (1, 4), (2, 2), (2, 1), (2, 4), (3, 3), (3, 4), (3, 2), (4, 3), (4, 2), (4, 1)\}$. Find the diagram of R , R^{-1} and R' . (CO1)

OR

(b) Define partial order relation. Is the relation of divisibility on the set $\mathbb{Z}$ of integers partial order relation? (CO1)

P. T. O.

3. (a) Define Hasse diagram. Consider the following Hasse diagram Fig. and the set $B = \{3, 4, 6\}$. Find, if they exist : (CO1)



- (i) Upper bound of B
- (ii) Lower bound of B
- (iii) Supremum (lub) of B
- (iv) Infimum (glb) of B

OR

- (b) By the principle of mathematical induction, prove that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n . (CO3)
4. (a) Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. (CO3)

OR

- (b) Show that $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction and $(p \Rightarrow q) \Leftrightarrow [(\sim p \vee q)]$ is a tautology. (CO3)

(3)

5. (a) Check the validity of the following argument : (CO3)

"If I study, then I will pass in examination. If I do not go to cinema, then I will study. But I failed in examination. Therefore, I went to cinema."

OR

- (b) Show that $\sim r$ is a valid conclusion from the premises
 $p \rightarrow \sim q, r \rightarrow p, q :$ (CO3)

- (i) with truth table
- (ii) without truth table